

Threat Intelligence Dashboard

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What I'm learned in this Program?

Topic

Computer Fundamentals

Networking

Programming & Scripting

Information Security Principles

Cryptography

Operating System Security

Network Security

Ethical Hacking & Penetration Testing

Web Application Security

Digital Forensics

Incident Response

Malware Analysis

Cloud Security

Risk Management & Governance

Cyber Law & Ethics

What I learn about in cybersecurity programm

Operating systems, hardware, file systems, and computer architecture.

TCP/IP, routing, switching, DNS, VPNs, wireless networks, and network protocols.

Basics of languages like Python, Java, C/C++, PowerShell, or Bash for automation and security tasks.

Confidentiality, Integrity, Availability (CIA Triad), authentication, authorization, and access control.

Encryption, hashing, digital signatures, certificates, and public key infrastructure (PKI).

Securing Windows and Linux systems, user permissions, logging, and hardening.

Firewalls, intrusion detection/prevention systems (IDS/IPS), network monitoring, and segmentation.

Finding vulnerabilities, reconnaissance, scanning, exploitation (in authorized environments), and reporting.

Common web vulnerabilities like SQL injection, cross-site scripting (XSS), and secure coding practices.

Recovering and analyzing digital evidence after incidents.

Detecting, investigating, containing, and recovering from cyberattacks.

Understanding viruses, ransomware, trojans, and basic malware behavior.

Securing cloud platforms, identity management, and cloud-specific risks.

Security policies, compliance, auditing, and managing organizational risk.

Privacy laws, regulations, ethics, and responsible security practices.

Project Purpose

A **Threat Intelligence Dashboard** is a cybersecurity project that collects and displays information about cyber threats from multiple sources in one place. Its purpose is to help security analysts monitor threats, identify malicious activities, analyze security risks, and respond quickly to cyberattacks. It improves decision-making by providing real-time insights through charts, alerts, and reports.

Project Objectives

- Developed a dashboard to collect and display cyber threat intelligence from multiple sources.
- Integrated threat intelligence APIs to retrieve information on malicious IPs, domains, URLs, and file hashes.
- Built interactive charts and tables to visualize threat trends and severity levels.
- Implemented search functionality to check the reputation of IP addresses, domains, and URLs.
- Added real-time alerts for newly detected high-risk threats.
- Stored and managed threat data in a database for analysis and reporting.
- Designed a user-friendly interface to help security analysts monitor and respond to cyber threats efficiently.

Structure of Your project

- 1.Requirement Analysis** – Identified the project's goals and selected reliable threat intelligence sources.
- 2.System Design** – Planned the dashboard architecture, database, and user interface.
- 3.Data Collection** – Retrieved threat data from APIs and public threat intelligence feeds.
- 4.Database Setup** – Stored threat information such as IPs, domains, URLs, and file hashes.
- 5.Backend Development** – Built the logic to fetch, process, and manage threat data.
- 6.Frontend Development** – Created a dashboard with charts, tables, search, and alerts.
- 7.Data Analysis** – Categorized threats based on type, severity, and risk level.
- 8.Testing** – Tested data accuracy, dashboard functionality, and performance.
- 9.Deployment** – Deployed the application for users to access and monitor cyber threats.
- 10.Maintenance** – Updated threat feeds and improved features based on user feedback.

Git Hub Link for Project

<https://github.com/harshilvadhiya/Harshil-vadhiya077.git>

Advantages Of Project

- Provides **real-time monitoring** of cyber threats.
 - Helps identify **malicious IPs, domains, URLs, and file hashes** quickly.
 - Improves **incident response** by detecting threats early.
 - Displays threat data using **easy-to-understand charts and dashboards**.
 - Saves time by **automating threat data collection** from multiple sources.
 - Supports **better decision-making** with accurate threat analysis.
 - Enhances the overall **security posture** of an organization.
 - Generates **reports and alerts** for security teams.
 - Stores historical data for **trend analysis** and future investigations.
 - Offers a **centralized platform** to manage and monitor cyber threat intelligence.
- Get smarter responses, upload files and images, and more.

Limitations of Project

- Depends on the **availability and accuracy** of external threat intelligence sources.
- Free APIs may have **rate limits** and limited threat data.
- Cannot detect **zero-day attacks** or unknown threats that are not yet reported.
- Requires a stable **internet connection** to fetch real-time data.
- Large volumes of threat data may affect **system performance**.
- May generate **false positives** or outdated threat information.
- Requires **regular updates and maintenance** to keep threat feeds current.
- Does not prevent attacks by itself; it only **monitors and reports** threats.
- Advanced features may require **paid threat intelligence services**.
- Effectiveness depends on **proper configuration and user expertise** in analyzing the results.

Challenges I'm faced in this project

- Integrating multiple **threat intelligence APIs** with different data formats.
- Handling **API rate limits** and authentication requirements.
- Cleaning and processing **large volumes of threat data**.
- Ensuring the dashboard displays **real-time updates** efficiently.
- Managing and storing data without affecting **performance**.
- Designing a **user-friendly and responsive interface**.
- Reducing **false positives** and duplicate threat records.
- Ensuring the **security** of the application and protecting API keys.
- Testing the system with different threat scenarios to verify accuracy.
- Keeping threat feeds and the dashboard **updated with the latest information**.

Conclusion

The **Threat Intelligence Dashboard** project successfully provides a centralized platform for monitoring and analyzing cyber threats in real time. It collects data from multiple threat intelligence sources and presents it in a simple and visual format for better understanding. This project helps in identifying malicious IPs, domains, URLs, and file hashes quickly, improving incident response and decision-making. It also demonstrates how cybersecurity and data visualization can be combined to support security operations. Overall, the project enhances awareness of cyber threats and shows the importance of threat intelligence in protecting systems and networks from potential attacks.

Thank you